Educational Sequence Mining for Dropout Prediction in MOOCs: Model Building, Evaluation, and Benchmarking

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***Abstract*—Due to the unprecedented growth in available data collected by e-learning platforms, including platforms used by massive open online course (MOOC) providers, important opportunities arise to structurally use these data for decision making and improvement of the educational offering. Student retention is a strategic task that can be supported by means of automated data-driven dropout prediction. Given the time-based nature of the collected data (user activity), these data can be viewed as sequences, and thus, sequence mining presents itself as a fitting set of techniques to automatically extract valuable insights. However, there is a lack of general guidelines for using sequence mining in specific educational settings, as well as little information on how different techniques perform in comparison to each other. We address these limitations with two main contributions. First, we propose a framework for applying sequence classification for dropout prediction in MOOCs. This framework includes two data-driven dropout definitions, the specification of data formatting and preparation tasks, and a blackprint on how to train dropout prediction models at suitable time points in the run of the course. Second, we conduct a benchmarking study of recent and well-performing sequence classification techniques, tested with different parametrizations on 47 real-life datasets from MOOCs, resulting in a comparative assessment of over 18 000 models. Our results provide insight into the performance differences between the techniques and allow us to formulate concrete recommendations toward the choice of suitable hyperparameters that have a significant influence on the predictive performance.**

***Index Terms*—Benchmarking study, dropout prediction, educational data mining, massive open online course (MOOC), sequence classification, sequence mining.**

I. Introduction

MASSIVE open online courses (MOOCs) have recently experienced a drastic surge in development and adoption by learners, making high-quality education widely accessible. However, since online education is mostly free or inexpensive and participants can leave MOOCs freely, these courses are typically characterized by high rates of student dropout. Completion rates of MOOCs have been shown to be as low as 13% [1]. This problem of student attrition in online education has been a point of concern of numerous educators and received wide attention in the literature [2], [3].

Most contemporary MOOC platforms generate increasing amounts of data, capturing every interaction between a student and the platform, such as watching a video, solving a quiz, or reading a textual explanation. These clickstream data, also referred to as tracking data or behavioral data, are typically recorded as a list of students' actions with corresponding timestamps. The availability of large amounts of data makes it possible to predict whether a certain student will not complete the course (i.e., drop out). These predictions can be valuable because well-timed and correct identification of student attrition allows teachers to retain students with targeted interventions and improve the quality of the offered content.

Sequence mining is a growing subdomain within data mining, which has found its application in a broad range of fields where data can be viewed as a sequence, including bioinformatics [4], web-page clickstream analysis [5], text categorization [6], ticket routing [7], and customer behavior analysis [8]. Similarly, student activity data can be viewed as sequences as well, and thus, sequential patterns can be extracted, as shown in [9] and [10].

Previous studies on using sequential patterns to predict dropout have employed process mining with success [11], [12]; however, the true power of sequence mining lies in dealing with a high number of items to which process mining techniques are typically not suited [10]. Due to the relative lack of works, it is unclear whether sequence classification algorithms are capable to grasp differences in student behavior in MOOC data in order to train performing dropout prediction models. Second, with the large variety in available sequence mining techniques, there is a lack of insight into which of those techniques would perform best for the task. Finally, there is an overall lack of guidance on how to deploy and parameterize sequence classification techniques for dropout prediction model building.

This study addresses these limitations with the following two contributions. The first contribution is a framework for applying sequence classification for dropout prediction in MOOCs, which is guided by the following research questions.

RQ1: What sequential techniques work best in the context of student dropout in MOOCs at different stages of the learning journey and how should they be parameterized?

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TABLE I
SEQUENCE DATABASE, CONSISTING OF A SET OF SEQUENCES WITH CORRESPONDING LABELS (TARGET VARIABLE)

ID	Sequence	Label
1	abbaac	0
2	abbabcca	0
3	aabacca	1
4	aabbcaacc	1

Possible interpretations of the items: a = watch video, b = answer question, c = read article, label = dropout.

RQ2: At what stage in the run of a MOOC does it become feasible to build predictive models? What gain can be expected by building models later in the course?

RQ3: What kinds of general, actionable, and data-driven definitions of student dropout in MOOCs can be proposed?

More specifically, our framework includes the formulation of different dropout definitions, data formatting and preparation, and analysis of appropriate prediction timing. The second contribution is a benchmarking study of recent and well-performing sequence classification techniques, tested with different parameterizations on 47 real-life datasets from MOOCs. During this study, over 18 000 models are built and analyzed, highlighting the differences in performance of certain techniques and leading to recommendations toward the choice of suitable parameters that influence the predictive performance.

The rest of the article is organized as follows. First, we discuss related work in Section II, including essential sequence mining concepts, sequence mining applications in education, the literature exploring MOOC data analysis, and research on dropout definitions. In Section III, we present a framework for applying sequence classification for dropout prediction, which includes data format and preparation, proposed dropout definitions, methods for handling class imbalance, evaluation metrics, and evaluation benchmark. Section IV provides results and recommendations, which are further summarized and discussed in Section V. Finally, Section VI concludes the article, outlines its limitations, and proposes directions for future work.

II. RELATED WORK

A. Sequence Mining: Essential Concepts

The sequential pattern mining problem was originally introduced by Agrawal and Srikant [13] as the task of finding all frequent sequential patterns existing in a sequence database. A sequence database SDB is a list of sequences, whereas a sequence is an ordered list of n item sets (or items) $s = < a_1, a_2, \ldots, a_n >$. An example of a sequence database is shown in Table I. A sequential pattern is called frequent if the number of sequences that contain this pattern is higher than a certain support level $\sup_{\min}$. The support level can be given as an absolute value, but it is often more convenient to represent it as a ratio or percentage of the sequences in the database that contain the pattern (sometimes also called the *relative support* [14]), which is used in this article. In the example in Table I, a sequential pattern $< a, b, b >$ is contained in 2 out of 4 sequences, which means that it is considered frequent if $\sup_{\min}$ is 0.5, and it is not frequent if $\sup_{\min}$ is 0.75.

In sequence classification, the task is no longer to discover all frequent sequential patterns, but to build a classifier based on a featurized version of a sequence, which is often achieved in two steps: 1) finding the most discriminative patterns and 2) consequently using them as features in a predictive algorithm. As such, the set of discovered patterns is different depending on which sequence classification approach is used (i.e., what kind of patterns are searched for and are considered to be the most discriminative by this algorithm). As different approaches tend to focus on mining different types of patterns, it can be expected that some of them are more suitable for certain types of data. For example, some techniques rely on n-grams, which consist of n consecutively occurring items, whereas others search for a co-occurrence of items at any place in a sequence, possibly with a gap in-between. However, it is also the case that, independently of the nature of the searched patterns, some sequence mining techniques are overall more computationally efficient and capable of producing more accurate predictive models based on any input data.

Sequence classification approaches can be indirect and direct. For the former, frequent patterns are first generated either for all sequences at once or separately for each class, and then used as features in a classifier, sometimes after applying feature selection to reduce the number of patterns. In the latter case, the patterns are sequentially retained according to their predictive quality, whereas the sequences are labeled based on these patterns within the same step, without subsequently applying a classifier. The majority of sequence classification techniques are indirect (e.g., SPADE and SCIS) or fall in-between direct and indirect approaches (e.g., iBCM), if they utilize the class label information to mine the most discriminative patterns. These techniques require a classifier to be used once the features are extracted. Some of the typical classification methods used in indirect approaches are support vector machines (SVM), random forest (RF), and Naive Bayes (NB).

B. Sequence Mining in Education

Sequence mining has been used for many purposes in learning analytics. It aims at extracting useful patterns that can inform a subsequent analysis. The general applicability of sequence mining techniques in the field of learning analytics was broadly discussed in [15] and mining sequential patterns has contributed to specific learning analytics tasks, such as predicting student success by analyzing transfer student records [16], generating personalized learning paths for different types of students [17], and recommending the most appropriate web-page to a student [18].

The range of sequence classification techniques used in these and other educational data mining studies is relatively diverse, for example, the GSP algorithm [17], [18], AprioriAll [18], PrefixSpan [18], SCIS [10], and cSPADE [10]. However, to the best of our knowledge, there are no studies that discuss how such techniques perform in comparison with each other.

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However, more recent and computationally tractable techniques such as BIDE (BI-Directional Extension) [19], MiSeRe (Mining Sequential Classification Rules) [20], and iBCM (interesting Behavioral Constraint Miner) [21] have not been applied on educational data yet.

Sequence classification aims at differentiating between two classes of learners. For instance, Kinnebrew et al. [9] compared patterns of students' productive versus unproductive learning behavior. In the context of dropout prediction, sequence mining is used to compare patterns of students who will drop out and those who will not. Such an application of sequence mining in the MOOC domain is discussed in [10]. The study suggests that sequence classification methods are capable to achieve high-performance levels when applied to behavioral data from MOOCs; however, with the large variety in available sequence mining techniques, there is a lack of insight into which of those techniques perform best for the task. Finally, there is an overall lack of guidance on how to deploy and parameterize sequence classification techniques for dropout prediction model building. This research gap is addressed by RQ1.

C. MOOC Data Analysis

Over the past few years, MOOCs have provided educational researchers with data on a nearly unprecedented scale, which has led to a vast surge in research in the field of learning analytics. Some studies explore general aspects related to MOOCs, such as educational theory, technology, individual case studies, participants, and providers, as described in a systematic literature review of MOOC-related research in [22]. However, the main focus of the research in the field of MOOCs pertains to the ongoing high dropout rates, which, depending on the course settings and dropout definitions, are reported to be as high as almost 90% [1].

Due to its high relevance and widespread occurrence, the problem of high dropout rates has received particular attention in the literature, being tackled both qualitatively (studying student engagement and methods for student attrition) and quantitatively (data-driven dropout prediction). In the case of the former, student engagement and its major causes are commonly investigated by using student interviews [2], [23], [24]. In the case of the latter, student data are analyzed to give insights into which students are likely to stop their course participation based on their prior actions [3], [25].

Recent advancements in data collection, storage, and analytics have led to an increase in quantitative studies of dropout prediction in MOOCs. The methods commonly used in such studies range from social network analytics [26], supervised learning with classification and regression [3], Hidden Markov Models [27], Ward's hierarchical and the k-means clustering methods [28], etc.

Some studies utilize their own rule-based predictors, e.g., [25], in which a prediction model is based on student activity patterns. The active mode predictor analyzes student activity to find patterns indicating a lack of motivation or ability, whereas the absence mode predictor counts the total number of days for which a student has been absent. More studies dedicated to dropout prediction in MOOCs are reviewed in [29], concluding that, among a variety of prediction features, clickstream data remain the most common. Other types of data commonly used for dropout prediction are forum posts, assignments, and sociodemographic data [30].

Gardner and Brooks [30] argue that feature extraction is one of the most difficult yet critical tasks for predictive modeling in MOOCs. Similarly, Taylor et al. [31] emphasize that the success of the model is attributed more to the variables used than to the model itself. Yet, the most common features constructed from clickstream data are mostly limited to counting-based activity features [30]. Such features, although relatively effective, only capture the intensity of the level of activity and miss some important information regarding the order of the clicks, which sequence classification is actually able to capture.

Another data analysis approach that can deal with time-stamped data without losing the information on the order and timing between events is process mining. Process mining has been used for MOOC data analysis, for example, to extract visual patterns that students follow during their MOOC trajectory in [32], which can be transferred into stochastic process models that can mimic different learners' behavior [33]. However, as discussed in [10], when it comes to predicting dropout in MOOCs, sequence mining is better capable of dealing with very granular behavioral data from a quantitative perspective, whereas process mining provides a better approach for the visualization of the differences between different student groups.

The influence of the timing of the prediction during the learners' experience on predictive accuracy is scarcely discussed in the literature. Most studies look into the timing dimension with the aim to optimize the retention of students. For example, Halawa et al. [25] incorporate the timing aspect by introducing an intervention window, which is the time between the first red flag the student receives and the last activity the student performed in the course. As such, they discuss that the first risk signals of the students appear 2 weeks before dropout; this, however, does not shed light on when it is best to run a predictive model for all students in the course at once as opposed to analyzing student dropout individually. Similarly, Tseng et al. [28] outline that the first 2 weeks form a critical point of time to retain students in MOOCs but do not investigate the point at which it is possible to accurately predict the students to be retained. This research gap is addressed by RQ2.

D. Dropout Definitions in MOOCs

To assess the likelihood of student dropout, it is important to have a clear and measurable definition of what a "dropout" is. However, since a major part of the studies explores the problem of student attrition with qualitative methods, their definitions of dropout are often rather descriptive and hard to measure. For instance, some papers classify learners into groups according to how engaged they are in the course. These classifications differ among the researchers and often depend on the objective of the research. Moreover, some studies only differentiate between two groups of students: ones who passed

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the course and ones who did not, whereas others report several types of learners with detailed descriptions. Some of such non-binary classifications are described ahead.

Kizilcec et al. [34] proposed four student groups according to their engagement style, which are "on track" (did assignments on time), "behind" (delayed assignments), "auditing" (did not do assignments but watched videos or took tests), and "out" (did not participate in the course at all). Additionally, the authors performed cluster analysis, revealing patterns of student engagement, such as "completing" (completed the majority of assessments), "auditing" (watched most of the videos but completed assessments infrequently), "disengaging" (completed assessments at the start of the course but withdrew engagement afterward) and "sampling" (merely explored some course videos). Similarly, in [35], five consistent engagement patterns were found across several MOOCs: "viewers" (mostly watched course lectures), "solvers" (mostly completed assignments), "all-rounders" (watched videos and completed assignments), "collectors" (downloaded videos without necessarily watching them), and "bystanders" (showed low activity in the course). In [36], on the other hand, a focus was put on students' intention in learning, which led to the division of students into "browsers" and "committed learners." The latter were further subdivided into "passive" participants (frequently watched lecture videos, did not participate much in the course forums, assignments, and quizzes), "active" participants (completed homework assignments, quizzes, exams, and peer-graded assessments), and "community contributors" (actively participated in the course, and showed specific interests in forum discussions and contributions to the public good). Finally, Tseng et al. [28] used cluster analysis to extract three groups of students: "active learners" (watched almost every video and handed in all assignments), "passive learners" ("only participate in some of the course's features") and "bystanders" (almost never handed in assignments nor watched any videos).

Some researchers define dropout as a student's last action (event) within a MOOC, where this action is considered to be watching a video [37], handing in an assignment or solving a quiz [31], writing on a forum [38], or simply the last activity [27]. Others define dropout as not obtaining a course certificate [39]. In [25], dropout is defined by two conditions: 1) being absent from the course for more than one month and 2) having viewed less than 50% of the videos. Finally, Whitehill et al. [40] labeled students as dropped out by the moment t if they 1) do not receive a certificate and 2) take no further actions between the moment t and the course-end date. The problem of obtaining a definition for data-driven dropout is addressed by RQ3.

III. Framework for Dropout Prediction Using Sequence Classification

This section presents a framework for predicting dropout in MOOCs using sequence classification, by discussing the format and preparation of the input data, dropout definitions, timing of predictions, and the class imbalance problem. In addition, it introduces evaluation metrics and evaluation benchmark.

TABLE II
Behavioral Data After Preprocessing the Native JSON File Used for Storage on the Edx Platform

Timestamp	Student	Activity	Chapter	Section
2020-11-20 13:25:36.844	Student1	enrol		
2020-11-21 01:31:58.687	Student2	play_video	C5	S3
2020-11-22 20:23:23.042	Student2	submit_task	C1	S5
2020-11-22 20:28:34.492	Student2	pause_video	C2	S3
2020-11-22 20:29:00.230	Student3	next_page	C3	S1
2020-11-22 20:30:48.177	Student3	view_forum	C2	S3

A. Data Format and Preparation

1) Input Data in a MOOC Context: Given the online-based format and nature of MOOCs, it is possible to capture and record students' actions performed within the online environment of the course. Most contemporary MOOC platforms are capable of providing such records of activity streams as users interact with the features of the platform. These behavioral data can provide insights into when and how students watch videos, start discussions on forums, answer exam questions, etc. While there are other types of data available in the MOOC context, such as demographic (age, gender, and degree), geographical (country of origin, native language), or psychological (goal striving, social belonging, and growth mindset), which as well might provide useful features for predictive analysis, in this work, we solely focus on behavioral data.

Behavioral data comprise a list of activities performed by a student, also referred to as log data or event log. The activities in MOOC log data represent fine-grained low-level actions, such as closing a web page, clicking a button, opening a new window, or tagging a selected text. Oftentimes, the data are available in structured file formats such as nested JSON (JavaScript Object Notation), which do require some adequate preprocessing before further analysis is possible. An example of such log data obtained after preprocessing the raw nested JSON data is shown in Table II.

2) Activity Granularity: The logging of actions offered by MOOCs is often done at a rather technical level, similar to raw web analytics data. In Table II, the *Activity* column represents a student action, whereas the *Chapter* and the *Section* columns indicate in which chapter and section of this chapter the action took place, respectively. In such tracking data generated by the edX platform, the low-level activities *NEXT PAGE*, *PLAY VIDEO*, or *VIEW FORUM* are rather meaningless for the analysis without information about the actual content element to which the action was applied. As such, appending content element information (in this case *Chapter* or *Section*), to action information (*Activity*) is expected to enhance the outcomes of analytics techniques.

In a previous study [10], we analyzed three levels of activity granularity: the one containing only the information about the nature of the action (*PLAY VIDEO*), and more detailed activity types with the Chapter (*PLAY VIDEO_C1*) and the Section (*PLAY VIDEO_C1_S2*) appended. The number of meaningful activity types in a typical MOOC for the least detailed level is ranging from 20 to 30. When the chapter information is introduced, this number grows to 100–150 for a 5-chapter course, whereas appending the section could potentially increase it to

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TABLE III
SEQUENTIAL DATA OBTAINED FROM THE DATA IN TABLE II

ID	Sequence	Label
1	enrol	1
2	play_video_C5_S3, submit_task_C1_S5, pause_video_C2_S3	0
3	next_page_C3_S1, view_forum_C2_S3	1

500–750 possible activities, assuming that each chapter contains 5 sections. However, in reality not all the activities are possible within each section, which means that the real number of distinct items for the most detailed activity granularity is normally no more than 500. This number of items is suitable to be handled by a typical sequence classification technique, as has been revealed by our analysis [10]. It is thus recommended to use all the available activity information for building sequences from log data.

3) Transforming Log Data to a Sequence Database: To make it possible to apply sequence classification to the data shown in Table II, certain transformations are required. More specifically, the data should contain sequences of student activities with a corresponding class label (dropout/nondropout). Each sequence represents a list of actions of one student in the order the actions were performed. An example of sequential data obtained after the transformation of the data from Table II is shown in Table III. The data for each student are transformed into a sequence labeled as 1 if we consider this student as a dropout and 0 if not. The difficulty here is to define what a "dropout" is, which is discussed in the next section.

B. Dropout Definitions and Timing of Dropout Prediction

The question of what is considered a dropout is not unanimously answered in the literature and strongly depends on a lot of factors, such as the way the course is designed, the preferences of the course instructors and, most importantly, the purpose of the analysis. For example, for some course instructors, it might be important that students participate in all the course activities timely, especially if a course has a limited and structured timeline, in which case, the absence from the course for 2–3 weeks is a reasonable dropout definition. For other courses, especially if they are self-paced, participation in a final exam or obtaining a course certificate is crucial, in which case not obtaining a certificate is considered a dropout case. Finally, some MOOC teachers might prefer that students watch at least 50% of available videos, and label them a "dropout" otherwise. The definitions proposed in the literature were discussed earlier in Section II-D.

All these definitions are reasonable and serve their individual purpose. However, a more general and objective dropout definition, preferably based on the data itself, would be preferable. Before outlining our dropout definition proposals, it is important to observe that the actionability of the derived model is a key problem for dropout prediction in MOOCs, in other words, whether the model is both sufficiently accurate to identify students at risk of dropping out as well as timely in the sense that the model should be potentially built as early as possible in the run of a course in order for retention efforts to have maximal effect. This entails the existence of a tradeoff between data availability and model actionability, as the former increases and the latter decreases as a course progresses. Second, it is crucial to prevent data leakage, meaning that the training data should not include information about a period after dropout. These problems are addressed with a window-based approach as follows.

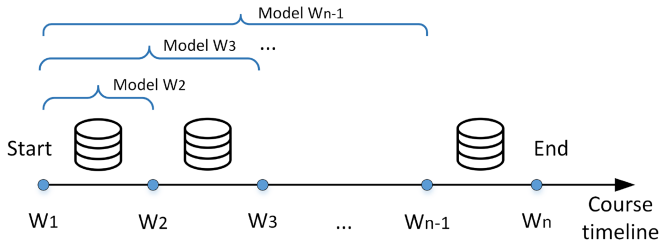


Fig. 1. Timing of prediction with a window-based approach.

- The data up to week W_i are used to predict whether a student will drop out in the next week W_{i+1} or in any of the following weeks (with $i = 1n - 1$, where n is the number of weeks of the course).

It is important for each dataset to only contain the data of students who have not dropped out before W_i, not only because it is more useful to detect the students who will drop out in the future but also to avoid data leakage.

In the absence of the information regarding teacher's preferences for a particular dropout definition, the only thing that is objectively known by the analyst is the last action of the student in the course. This end of course participation by itself might not imply that the course was not completed by the student, especially if the last event has happened after the last week of the course. However, this information can be used to define a *local dropout*:

Definition 1 (Local dropout): A student is considered a dropout in week i, if the last action of the student was performed in week W_i, and therefore, no more activity has happened after W_i.

From the point of view of model actionability, if we are to build a predictive model at the beginning of W_i and we use all the available data of W_{i-1}, it is appealing to predict the students who will likely drop out in the next week, as those are the students we would like to target primarily. Thus, the idea of predicting local dropout is to use the data of week W_i to predict whether a student will drop out in W_{i+1}, as illustrated in Fig. 1.

This notion of local dropout by definition includes students who were absent from the course for a certain time, as well as students who did not participate in a final exam, unless this exam has happened in the week before the moment of prediction but then building a model at this moment is no longer relevant. An additional advantage of the local definition is that it helps to prioritize the students that need to be retained first as they are going to drop out sooner.

However, despite its attractiveness, local dropout might have important implications for building dropout prediction models. This is because students dropping out later than week W_i will not be considered as dropouts, and thereby mixing students who will drop out later with students who do not. This

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can significantly impact the performance of predictive models, given that achieving class separability will be much more challenging. Therefore, we also propose a second definition, *global dropout*, which is formulated as follows:

Definition 2 (Global dropout): Students are labeled as "dropout" if their last action in the course happens before the moment when the last chapter of the meaningful content (i.e., content that is part of the final exam/quiz/assignment) is revealed, or when they do not appear on the platform for more than T weeks (even if they still reappear afterward), with $T = \mathrm{round}(N/3)$, where N is the number of weeks of the course.

For example, for a 5-week course $T = 2$, whereas for an 8-week course $T = 3$. The reasoning behind this definition is the following. First, we approach dropout as "stop participating in the course" and do not link it to a final score, which makes this definition suitable for courses with different requirements regarding earning certificates. For example, some courses may only give certificates to students who pay an extra fee, and thus models trained on such a target variable will likely be biased. Second, if a student never returns to the platform after the last content was revealed, it is reasonable to assume that this student is not capable of fulfilling all the course requirements. Additionally, a long absence of a student is a well-grounded argument for labeling the case as "dropout," and it has been repeatedly used in the literature. For example, as discussed in Section II-D, one of the conditions for dropout in [25] was an absence of a student for one month. As the time span of one month is calculated for a specific course, we take a step further and define the absence time as one-third of the number of weeks of the course, which is found empirically optimal by our analysis of seven courses with the length ranging from 5 to 8 weeks.

In comparison with local dropout, the definition of global dropout does not only contain students who drop out in week i, but also students who drop out in the weeks after that. The students who dropped out before the prediction moment are not included in the training set.

It is important to note that the proposed local and global dropout definitions are only applicable for MOOCs with a certain time structure, which are the majority of MOOCs. Such a time structure includes start time, end time, deadlines for assignments, and defined moments of revealing pieces of content. In case of courses without a timeline (self-paced courses), defining dropout based on stopping participation in a certain week is simply not useful, as students have their own relative timeline, start and finish the course at different time moments, and also spend different amounts of time to complete the course. Thus, a potential dropout definition in a self-paced course is mostly reliant on the need of the course instructor. One of the ways to define a global dropout in a self-paced course is to use the information whether a student participated in the final exam.

C. Handling Class Imbalance

The majority of classification algorithms work well when the class distribution of the target variable is balanced, meaning that the number of instances of each class is comparable. In case of dropout prediction, the number of dropout cases is typically significantly larger than the number of nondropout cases, which might affect the performance of classifiers. If a window-based approach is used wherein only the students that drop out in the next week are considered, the class distribution becomes reverse as the number of next-week dropouts is relatively small compared to the whole course population. Such datasets with an unequal proportion of instances in each class are called imbalanced.

The data-level solutions to class imbalance comprise various resampling techniques that change class distribution by increasing the minority class observations or decreasing the majority class instances [41]. The two most basic sampling techniques are random oversampling (i.e., randomly duplicating minority class examples) and random undersampling (i.e., randomly discarding majority class examples). Examples of more complex resampling techniques are the synthetic minority oversampling technique (SMOTE) [42] and the adaptive synthetic sampling approach (ADASYN) [43]. While having their advantages, data-level methods may suffer from some drawbacks, which are widely discussed and addressed in the literature [41], [44], [45]. Given the scope of this work, which is to compare available sequence classification techniques "as they are," without modifications, we do not look into more powerful, but also more complex ways to solve data imbalance problems, such as algorithm-level techniques or cost-sensitive methods. Given the effectiveness of data-level methods, as well as the fact that they are easy to implement, in this work, we analyze the impact of different approaches to oversampling the minority class on the performance of the sequence classification techniques.

D. Evaluation Metrics and Their Interpretation in a MOOC Context

The model performance can be evaluated using the most common classification performance metrics including accuracy, precision, recall, area under the curve (AUC), F1-score, balanced accuracy, and top decile lift. Accuracy (i.e., the percentage of correct predictions) is the most commonly reported metric in the sequence classification domain, and the most easily interpretable. However, particularly for imbalanced setups, accuracy is often an overly optimistic metric to compare predictive performance. Therefore, it is worthwhile to look into other metrics, such as balanced accuracy or AUC (area under the ROC curve). AUC is an often used metric for classification performance comparison, as it measures the tradeoff between the true and false positive rates over different classification thresholds. While AUC is an excellent aggregate comparative metric, other metrics have a stronger relation to the actual problem domain. Precision is such a metric, as it addresses the question "What is the percentage of students that was actually going to dropout out of the students that were predicted as such?." This could be a useful metric to look at if a course instructor is interested to know how effective a potential targeted intervention will be and how many of the potential

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TABLE IV
DESCRIPTIVE STATISTICS OF THE LOCAL DROPOUT DATASETS

Course/duration	Subset	# of students	% of dropout	# of activities	# of events
EUHURIx2015 6 weeks	W2	1439	21	174	69805
	W3	1422	10	207	113010
	W4	1360	10	220	140060
	W5	1288	10	254	176729
EUHURIx2016 6 weeks	W2	549	15	197	28496
	W3	557	10	242	42829
	W4	546	10	280	59085
	W5	526	9	337	76252
GRAPHx2015 8 weeks	W2	792	17	244	42428
	W3	754	13	261	60728
	W4	701	14	286	74956
	W5	632	12	317	98604
	W6	584	16	348	110667
	W7	520	16	380	120960
GRAPHx2017 8 weeks	W2	274	15	361	18048
	W3	276	12	408	27921
	W4	281	9	462	36737
	W5	282	16	512	47871
	W6	256	12	560	55605
	W7	238	14	610	63627
PSYx2015 5 weeks	W2	642	19	271	48875
	W3	586	24	380	90533
	W4	482	24	506	121405
PSYx2016 5 weeks	W2	84	23	295	7277
	W3	96	32	412	13436
	W4	84	26	503	18238

TABLE V
DESCRIPTIVE STATISTICS OF THE GLOBAL DROPOUT DATASETS

Course	Subset	# of students	% of dropout	# of activities	# of events
EUHURIx2015	B	1520	83	180	82464
	M	1422	81	207	113010
	E	1360	79	220	140060
EUHURIx2016	B	549	83	197	28496
	M	557	82	242	42829
	E	546	80	280	59085
PSYx2015	B	642	79	271	48875
	M	586	74	380	90533
	E	482	71	506	121405
PSYx2016	B	87	85	295	7422
	M	99	84	412	13880
	E	87	75	504	18907
GRAPHx2015	B	792	89	244	42248
	M	701	85	286	74956
	E	584	82	348	110667
GRAPHx2017	B	274	89	361	18048
	M	281	86	462	36737
	E	256	83	560	55605
UMLx2017	B	1209	75	244	233098
	M	1028	73	260	242348
	E	784	71	249	201884

nondropout cases it would involve. Recall, on the other hand, addresses the question of "What percentage of the students that were going to drop out was successfully detected by the predictive algorithm?," which is useful to investigate if the goal is to retain as many students as possible. F1-score is a harmonic mean between these two measures. Finally, another very actionable metric is top decile lift. It measures how much better a classification model is, compared to a random selection, considering the top 10% predicted instances with the highest prediction probability. Given that, typically, resources that are strongly constrained in educational settings and models that are better at identifying the most likely students at risk of dropping out are preferable. Top decile lift exactly measures this aspect.

E. Experimental Benchmark

1) Data Description: This study analyzes data from seven MOOCs from the EdX platform, abbreviated according to their name on edX as EUHURIx2015, EUHURIx2016, ePSYx2015, ePSYx2016, GRAPHx2015x, GRAPHx2017x, and UMLx2017. All courses with the exception of UMLx2017 are week-based and have a strict timeline, with a course duration ranging from 5 to 8 weeks. In order to compare different dropout definitions, as well as data processing methods and prediction timings, the data from these MOOCs are processed in line with the description in Section III in two distinct ways: using 1) local and 2) global dropout.

In case of local dropout (see Table V), all available data are divided into smaller subsets, containing only the data up to a certain week, with a target variable being students who dropped out during the next week after the prediction moment. In case of global dropout (see Table IV), the target variable is defined at once for the whole dataset and stays the same for each subset. However, similarly to the case of local dropout, the data are divided into smaller subsets, in order to assess the quantity of data and the timing of prediction needed to build an insightful model. The subsets for global dropout are marked as the beginning (B), middle (M), and end (E) of the course, and contain data up to one-third of the course, half of the course, and two-thirds of the course, respectively.

Finally, the last course, UMLx2017, is a self-paced course with a mandatory final exam. As this course is not time-based, we adapt a dropout definition based on the course instructor preferences and define dropout as not participating in the final exam. To avoid data leakage, the activity that corresponds to participation in the final exam has been removed from the subsets. Moreover, in order to evaluate dropout prediction performance at different intervals, we only considered the period of the course with the highest activity, given that despite being self-paced, timing was roughly determined by the final exam date. As such, a total of 47 datasets from seven MOOCs are used. Tables V and IV summarize the main characteristics of the datasets, as follows.

1) The column *Course/duration* contains the abbreviation of the course, the year it was run, and its duration. The names of the MOOCs are *EUHURIx The EU and Human Rights*, *GRAPHx The Great War and Modern Philosophy*, *PSYx Trends in e-Psychology*, and *UML Class Diagrams for Software Engineering*.
2) *Subset* indicates the moment of prediction and a corresponding subset constructed with the data up to this moment. In case of weekly subsets, we build predictive models for each week, starting from week 2 and up to the second-to-last week. Thus, for a course that runs for 6 weeks, 4 models are obtained: in the beginning of weeks 2, 3, 4, and 5. Accordingly, only the data up to weeks 2, 3, 4, and 5 are used for each subset. To avoid data leakage, the data from the students that stopped the course participation before the prediction moment are removed from the subsets. The students that are labeled as dropouts are those who stop their course participation within a week after the prediction moment. For

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TABLE VI
OVERVIEW OF THE SEQUENCE CLASSIFICATION TECHNIQUES ANALYZED IN THIS STUDY

Technique	Description	Hyperparameters
cSPADE	Based on window, gap, length, width, and other constraints. Uses vertical id-list database format and two different search strategies: breadth-first and depth-first.	Support
PrefixSpan	Efficient sequence mining approach based on growing a pattern database of prefix projections.	Support
BIDE	Closed sequence mining approach, uses the BackScan pruning method and the ScanSkip optimisation procedure.	Support
MiSeRe	Randomly generates diverse sequences and applies a Bayesian approach to retain interesting sequences.	Maximum runtime (in seconds)
SCIS	Extension to SPADE based on an interestingness measure that next to the support of a sequence also consists of the proximity of its items.	Support, interestingness, maximum length of sequences
iBCM	Relies on mining a set of behavioural constraints based on Declare language.	Support

example, for the subset W3 the data of weeks 1 and 2 are used to predict which students will dropout during week 3.

3) *# of students* denotes the number of students in each subset after removing the students that stopped the course participation before the prediction moment. It is important to note that sometimes the number of students in the following weeks increases because certain students join the course later than its beginning.
4) *% of dropout* is the percentage of students that drop out from the course within a week after the prediction moment. They are the students we are trying to detect (i.e., those for which the target variable is set to 1).
5) *# of activities* is the number of distinct student activities of the most fine-granular level. In other words, this is the number of possible items in the sequential database.
6) *# of events* denotes the total number of activities conducted by the students in the subset.

F. Sequence Classification Techniques and Associated Challenges

An overview of the included techniques is presented in Table VI. All the techniques are available as part of the SPMF open-source Java library [46], except for iBCM for which an implementation is made available on Github [47].

When featurizing data, we have faced two challenges. First of all, the way the algorithms had been implemented only allowed us to search for sequential patterns in a complete dataset, which made implementing a training/test split or a cross-validation setup for a proper model validation an additional challenge. As an important part of solving a machine learning problem is its valid evaluation of unseen data, featurizing all available data and then feeding these features to a classifier leads to data leakage and overfitting. In other words, this leads to building overly accurate models, which will be useless in real life. To the best of our knowledge, this problem is not directly assessed in sequence mining research. Moreover, also in educational data mining literature, many studies pay too little attention to guaranteeing proper evaluation. In order to be able to apply the aforementioned sequence classification techniques to the data from MOOCs, they first had to be adapted so that they are capable to first search for discriminative patterns in the training set, and then featurize the test set with the same patterns. This allows us to avoid the leakage of the label information of the test set and thus be able to estimate the real-life performance of the model.

The second challenge is linked to the computational complexity of the task at hand. The experiments were executed on an Intel Xeon CPU E5-2699 (v3) with 256 GB of RAM. All support-based algorithms were prevented from running longer than 60 min to generate sequential patterns (except for MiSeRe, which takes running time as input parameter). The techniques that run out of time during the feature generation process or were not capable of generating features for a particular dataset, have been assigned to produce majority class prediction models. For support-based techniques, namely cSPADE, PrefixSpan, BIDE, iBCM, and SCIS, after an initial investigation of support levels at intervals of size 0.1, a decision was made to report results at three levels of support: 1) 0.4, 2) 0.6, and) 0.8. Please note that running times at lower support levels was often exceeding reasonable time spans given the large sets of features being generated. Moreover, given the complexity of the datasets, with hundreds of items in the sequence alphabet, it is not surprising that even such well-performing techniques as the selected ones were not all capable to complete the featurization task within the set limit of 60 min. This correlates with the fact that the number of produced patterns for the chosen techniques varied significantly, from dozens for iBCM to hundreds of thousands for PrefixSpan. As PrefixSpan and SCIS were not capable to featurize the vast majority of the datasets within the defined timeframe, they are not included in the results presented ahead.

For the remaining techniques, a 10-fold cross-validation setup is used, in which the features are produced 10 times for a 9/10 of the datasets and the remaining 1/10 of the data is used as a test set. The reported metrics are averaged across all folds. In total, the features are generated 30 times for each of the 47 datasets: for support values of 0.4, 0.6, and 0.8 in case of the support-based techniques and with 1, 2, and 5 s run times for MiSeRe.

Furthermore, we build predictive models from the generated features using four classifiers: 1) SVM, 2) NB, 3) RF, and 4) logistic regression (LR), for which the Python library *scikit-learn* was used [48].

As discussed in this section, the MOOC data suffer from class imbalance. To assess the influence of class imbalance handling methods on the models' accuracy, all the experiments are performed four times: once without any class imbalance countermeasure in addition to using simple majority class oversampling, SMOTE, and ADASYN.

Finally, since the number of generated sequential patterns is very large at times, we utilize principal component analysis (PCA) as a method for dimensionality reduction. All the models are built for unprocessed features first and then reduced to the 10 principal components of the features. This allows us to

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TABLE VII
AVERAGE AUC AND TOP DECILE LIFT FOR THE BEST RESULT PER DATASET FOR ALL SEQUENCE CLASSIFICATION TECHNIQUES AND A BAG-OF-WORDS MODEL

Technique	AUC	Lift
iBCM	0.71	2.43
SPADE	0.66	2.23
MiSeRe	0.63	2.16
BIDE	0.64	1.68
BOW	0.61	1.92

analyze the influence of a reduced feature space on the predictive power of the model.

To summarize, the final results are reported based on 18 048 experiments, obtained by combining 47 datasets, 4 sequence classification techniques, 3 support levels/running times, 4 downstream classifiers, 4 class imbalance methods, and 2 dimensionality reduction approaches.

IV. RESULTS

A. Baseline Comparison

To compare the results of the sequence classification approach, we have compared them with the performance of a bag-of-words model [49], containing a binary feature indicating the presence or absence of an activity in a sequence. The results of the comparison are shown in Table VII. As can be seen in Table VII, the bag-of-words on average outperformed all the sequence classification techniques, if optimal hyperparameters are chosen. Those optimal hyperparameters are further discussed in the remainder of this section.

B. Local Versus Global Dropout Prediction

First, we analyze how the definition of the target variable influences the predictive performance. Fig. 2 shows that the distributions of both AUC and lift are significantly more favorable in case of predicting dropout globally. The overall average AUC for global dropout is 0.66, which is a positive result, considering that this is the average result for all trained models across all datasets. Moreover, best performing models reach an AUC of 0.9 and higher. On the contrary, the distribution of the metrics for local dropout indicates that even though a few well-performing models have been obtained, the overall quality of the models remains close to random guessing.

Two conclusions could be made when interpreting these results from an educational viewpoint. First of all, the well-performing global dropout prediction models allow us to conclude that dropped-out students behave somewhat differently from those completing the course, and this difference can be captured by mining sequences of their actions. For instance, it might be that students who start with answering quizzes instead of reading theory first and show lower levels of performance. Furthermore, these behavioral differences appear to occur throughout the whole course and not only a week before the students stop their participation, which is why local dropout is much harder to predict, a finding that is not completely unexpected.

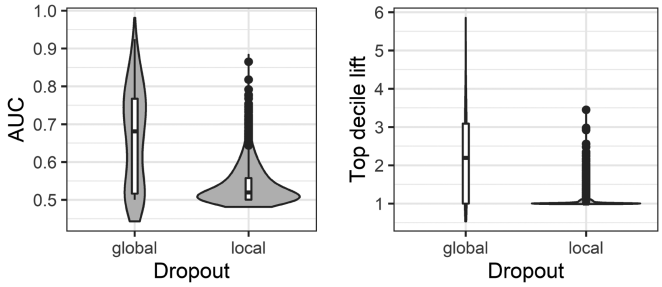


Fig. 2. Distribution of AUC and top decile lift for global and local dropout (summarized across 18 048 experiments).

Accordingly, even though it would be an interesting capability to predict those students who drop out in the next week, it is more realistic to build models to detect all students who will not complete the course. As such, in the following, we discuss the models for global dropout only (a total of 8064 experiments).

C. Downstream Classifier Configuration: Algorithm, Dimensionality Reduction, and Class Imbalance

The performance of the downstream classification algorithms in terms of AUC and top decile lift, averaged across all the datasets and sequence mining techniques, class imbalance methods, and dimensionality reduction approaches is summarized in Table VIII and Fig. 3. Since the results are averaged across all 21 datasets and all 4 sequence classification techniques, the variations in performance between classifiers are not seemingly high. However, even a slight increase in average AUC and lift values can indicate a significantly better performance for some models.

Overall, a few tendencies can be observed. First of all, LR models tend to show better results, especially when the top decile lift is considered. For LR, the average lift is 2.38, higher than the overall average of 2.20. Next, it can be noted that for LR and NB, the influence of reducing the number of features using principal components is positive, whereas this is not the case for RF and SVM. This result is in line with the fact that LR and NB can cope worse with high dimensionality, while SVM and RF are more robust against a larger feature space. Please observe that a disadvantage of PCA in educational sequence mining is that it hinders potential feature interpretation. However, it is similarly challenging to interpret an RF model. Based on the observed results, our recommendation is to use an LR combined with regularization, feature selection, or a dimensionality reduction method, depending on whether an easier interpretation is desirable. A proper feature selection technique is especially important when a huge feature space is generated by some of the sequence classification techniques. RF can be applied without a feature selection method, as tree-based models naturally rank features by how well they improve the purity of the nodes. In addition, SVM shows a good level of performance without feature reduction as well, whereas NB is generally outperformed by other classifiers.

Among the oversampling methods, SMOTE shows the best performance in terms of top decile lift, whereas random

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TABLE VIII
AVERAGE AUC AND TOP DECILE LIFT FOR DIFFERENT COMBINATIONS OF CLASSIFIERS, CLASS IMBALANCE METHODS, AND DIMENSIONALITY REDUCTION APPROACHES (SUMMARIZED OVER ALL GLOBAL DROPOUT DATASETS, A TOTAL OF 8064 EXPERIMENTS)

Classifier	Imbalance	Dimensionality	AUC	Lift
Logistic regression	None	None	0.67	2.41
Logistic regression	None	PCA (n=10)	**0.68**	**2.45**
Logistic regression	Oversampling	None	0.66	2.35
Logistic regression	Oversampling	PCA (n=10)	**0.68**	2.43
Logistic regression	ADASYN	None	0.65	2.27
Logistic regression	ADASYN	PCA (n=10)	0.67	2.36
Logistic regression	SMOTE	None	0.66	2.31
Logistic regression	SMOTE	PCA (n=10)	0.67	2.43
Logistic regression			0.67	2.38
Naive Bayes	None	None	0.66	1.66
Naive Bayes	None	PCA (n=10)	**0.67**	**2.32**
Naive Bayes	Oversampling	None	**0.67**	1.68
Naive Bayes	Oversampling	PCA (n=10)	0.66	2.13
Naive Bayes	ADASYN	None	0.66	1.70
Naive Bayes	ADASYN	PCA (n=10)	0.66	2.23
Naive Bayes	SMOTE	None	0.66	1.69
Naive Bayes	SMOTE	PCA (n=10)	0.66	2.21
Naive Bayes			0.66	1.95
Random forest	None	None	**0.68**	**2.38**
Random forest	None	PCA (n=10)	0.67	2.31
Random forest	Oversampling	None	0.67	2.25
Random forest	Oversampling	PCA (n=10)	0.66	2.20
Random forest	ADASYN	None	0.66	2.19
Random forest	ADASYN	PCA (n=10)	0.65	2.13
Random forest	SMOTE	None	0.66	2.24
Random forest	SMOTE	PCA (n=10)	0.66	2.20
Random forest			0.66	2.24
SVM	None	None	0.64	**2.40**
SVM	None	PCA (n=10)	0.64	2.20
SVM	Oversampling	None	**0.68**	2.38
SVM	Oversampling	PCA (n=10)	0.66	2.13
SVM	ADASYN	None	0.67	2.32
SVM	ADASYN	PCA (n=10)	0.65	1.96
SVM	SMOTE	None	**0.68**	2.38
SVM	SMOTE	PCA (n=10)	0.66	2.11
SVM			0.66	2.23

The best results are shown in bold.

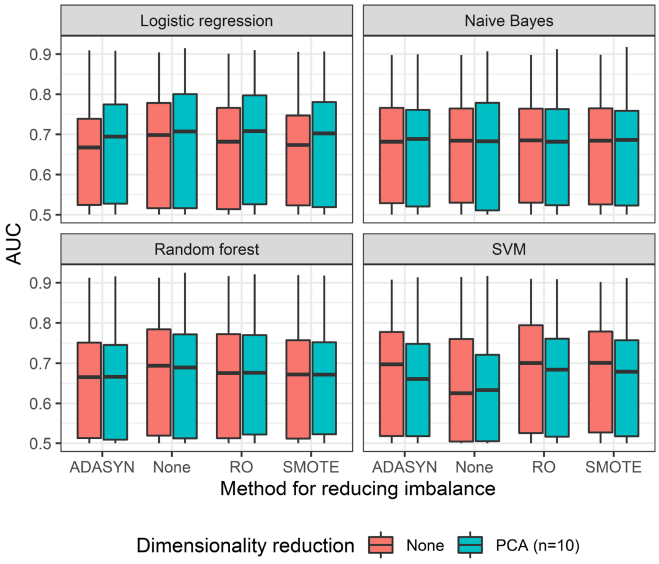


Fig. 3. Distribution of AUC for different combinations of classifier, method for imbalance, and PCA (global dropout, 8064 experiments).

oversampling shows better AUC. Generally, SMOTE and ADASYN help to avoid overfitting when compared to random oversampling, since the newly generated minority class observations are not copies of the minority class, but synthetically

TABLE IX
AVERAGE AUC AND TOP DECILE LIFT FOR DIFFERENT COMBINATIONS OF SEQUENCE CLASSIFICATION TECHNIQUES AND DIMENSIONALITY REDUCTION APPROACHES (GLOBAL DROPOUT, RANDOM OVERSAMPLING, 2016 EXPERIMENTS)

Technique	Dimensionality	Average # of features	AUC	Lift
BIDE	None	6.3	0.56	1.29
BIDE	PCA (n=10)	6.3	0.56	1.29
iBCM	None	1646.87	0.74	2.55
iBCM	PCA (n=10)	1646.87	0.74	2.63
MiSeRe	None	95.1	0.73	2.84
MiSeRe	PCA (n=10)	95.1	0.73	2.93
SPADE	None	398.1	0.65	1.98
SPADE	PCA (n=10)	398.1	0.65	2.04

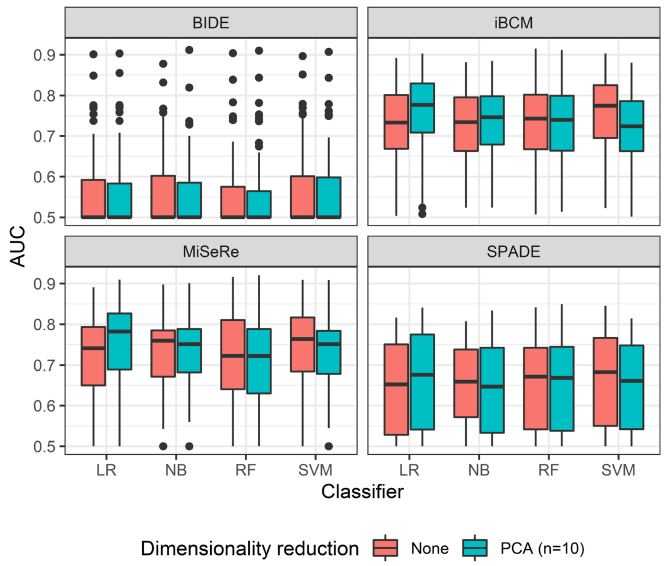


Fig. 4. Distribution of AUC for different combinations of technique, classifier, and PCA (global dropout, random oversampling, 2016 experiments).

generated data points. However, for binary features generated by sequence mining algorithms, the meaning of synthetically generated features can be lost as they are forced to deal with binary 0/1 features as numerical values. When it comes to a method for handling class imbalance, our recommendation is to use random oversampling, since it works better with binary features [50].

D. Choice of Classifier and Dimensionality Reduction Depending on the Sequence Classification Technique

Given that the choice of oversampling technique has little connection with classifier and dimensionality reduction, and given the result obtained in the previous section that random oversampling is more suitable for binary features compared to other methods, we now only look into models built using random oversampling. The remaining 2016 experiments are analyzed to investigate whether the choice for PCA as feature reduction could potentially depend on the sequence classification technique. Table IX indicates that, on the one hand, in line with expectations, the lift of models with PCA is higher for techniques that generate larger amounts of features. On the other hand, AUC stays the same for all techniques, irrespective

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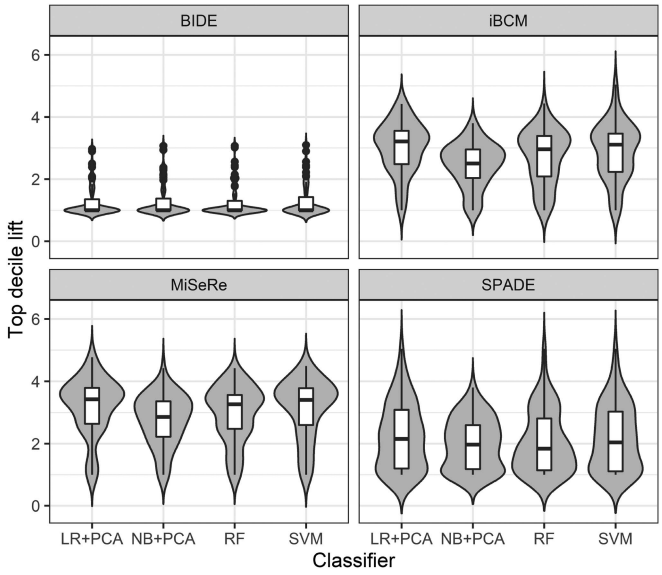


Fig. 5. Distribution of top decile lift for different combinations of technique, PCA, and classifier (global dropout, random oversampling, 1008 experiments).

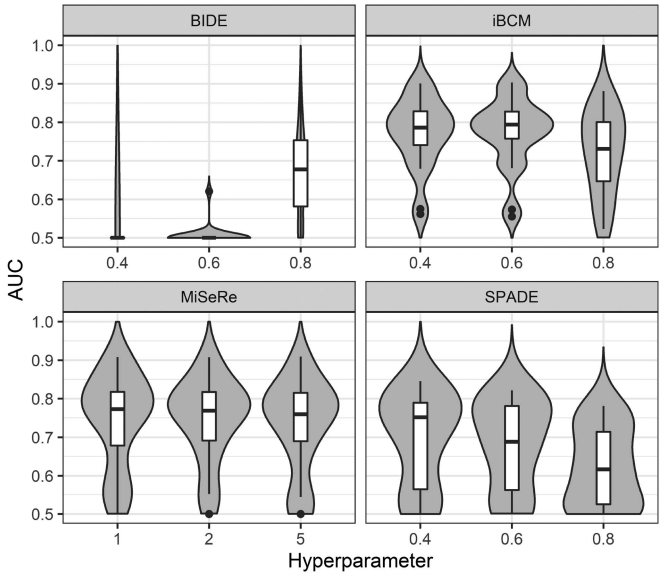


Fig. 6. Distribution of AUC for each technique for different hyperparameters: Support for BIDE, SPADE, and iBCM, and run time in seconds for MiSeRe (global dropout, random oversampling, no PCA + SVM, 252 experiments).

TABLE X
AVERAGE AUC AND TOP DECILE LIFT FOR DIFFERENT COMBINATIONS OF TECHNIQUE, PCA AND CLASSIFIER (GLOBAL DROPOUT, RANDOM OVERSAMPLING, 1008 EXPERIMENTS)

technique	classifier	AUC	Lift
iBCM	LR+PCA	0.754	3.01
iBCM	SVM	0.754	2.87
MiSeRe	LR+PCA	0.746	3.20
MiSeRe	SVM	0.741	3.12
iBCM	RF	0.735	2.70
iBCM	NB+PCA	0.731	2.45
MiSeRe	NB+PCA	0.724	2.76
MiSeRe	RF	0.718	2.95
SPADE	SVM	0.663	2.17
SPADE	LR+PCA	0.658	2.21
SPADE	RF	0.647	2.03
SPADE	NB+PCA	0.64	1.98
BIDE	SVM	0.558	1.29
BIDE	LR+PCA	0.557	1.28
BIDE	RF	0.555	1.28
BIDE	NB+PCA	0.55	1.29

The results are sorted in the descending order based on AUC.

the use of PCA. As shown in Fig. 4 and in the previous section, the choice of using dimensionality reduction is much more correlated with the choice of the classifier, and the pattern described in the previous section is the same for all techniques.

Additionally, Fig. 4 illustrates a significant difference between the performance of different sequence classification techniques. It is clear from the figure that BIDE is outperformed by SPADE, whereas SPADE is outperformed by both iBCM and MiSeRe.

Given the previous conclusion that NB and LR work best with PCA, whereas RF and SVM are more suitable to use without PCA, we now analyze whether a combination of classifiers and their preferable feature selection perform similarly. Fig. 5 and Table X show the performance of different sequence classification techniques in terms of top decile lift, random oversampling, and global dropout. Similarly to the results from Table VIII, it can be observed that NB might not be the best choice of classifier with any of the analyzed sequence classification techniques. The other three classifiers show comparable results, with SVM and LR performing slightly better than RF. This tendency is clear for iBCM, MiSeRe, and SPADE, and perhaps less clear for BIDE, most likely due to an overall lower performance of this technique.

Given that SVM is shown to obtain consistently good results for all techniques even without PCA, in the following section, we present the results only for the models with SVM, which amounts to the remaining 252 experiments (global dropout, no PCA, random oversampling, SVMs). It is important to note that this does not imply that SVM is always better than RF and LR; in fact, all three techniques show similar performance, as discussed in the previous section.

E. Influence of the Hyperparameters of the Sequence Classification Techniques

By definition, support influences the number of patterns mined by a sequence classification algorithm. In practice, this means that if it is set too low, the number of patterns that satisfy the pattern frequency criteria is simply too large, which leads to exceedingly long computational time and/or incapability of completing the search. In addition, an excessively large number of features fed to a predictive model without an appropriate feature selection step might lead to overfitting or make the model overly complicated. Therefore, a low support level such as 0.1 is rarely a good choice for the majority of sequence classification techniques. However, when the support level is too high, there is a chance of missing relevant and

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TABLE XI
AUC AND TOP DECILE LIFT FOR EACH COMBINATION OF TECHNIQUE, COURSE, AND TIMING OF PREDICTION (84 EXPERIMENTS, GLOBAL DROPOUT, SVM, RANDOM OVERSAMPLING, NO PCA, OPTIMAL SUPPORT/RUN TIME)

		BIDE		iBCM		MiSeRe		SPADE	
Course	Timing	AUC	Lift	AUC	Lift	AUC	Lift	AUC	Lift
EUHURI2015	B	0.70	1.68	0.76	2.26	0.79	2.73	0.76	1.97
EUHURI2015	M	0.76	1.88	0.78	2.50	0.80	2.76	0.79	2.50
EUHURI2015	E	**0.77**	**2.11**	**0.83**	**3.11**	**0.84**	**3.40**	**0.82**	**2.97**
EUHURI2016	B	0.50	1.04	0.70	2.48	0.62	1.56	0.70	2.54
EUHURI2016	M	0.67	1.54	**0.76**	**3.36**	0.74	3.36	**0.76**	3.42
EUHURI2016	E	**0.69**	**1.84**	**0.76**	3.32	**0.75**	**3.73**	0.75	**3.88**
GRAPHx2015	B	0.58	1.31	0.69	1.91	0.68	2.18	0.68	2.52
GRAPHx2015	M	0.68	1.84	0.80	**3.95**	**0.80**	3.54	0.78	3.68
GRAPHx2015	E	**0.75**	**2.22**	**0.80**	3.30	0.82	**4.33**	**0.80**	**4.22**
GRAPHx2017	B	0.65	**2.56**	0.79	**4.77**	0.77	**4.42**	**0.82**	**5.04**
GRAPHx2017	M	**0.68**	2.11	0.80	3.37	0.75	4.14	0.65	3.37
GRAPHx2017	E	0.64	2.44	**0.81**	3.37	**0.82**	3.78	0.50	1.00
PSYx2015	B	0.50	1.00	0.77	**3.54**	0.76	3.35	0.75	2.93
PSYx2015	M	0.75	1.91	0.83	3.52	**0.85**	**3.71**	0.84	**3.59**
PSYx2015	E	**0.78**	**2.09**	**0.84**	3.28	0.56	2.60	**0.85**	3.35
PSYx2016	B	0.51	1.00	0.57	1.47	**0.56**	1.47	0.55	2.21
PSYx2016	M	**0.52**	1.00	**0.68**	**1.86**	0.54	**3.09**	**0.71**	**3.09**
PSYx2016	E	0.50	1.00	0.56	1.00	0.50	1.00	0.57	1.30
UML2017	B	0.85	**3.10**	0.88	**3.98**	0.89	**3.98**	**0.55**	**2.82**
UML2017	M	0.65	2.39	**0.90**	3.67	0.90	3.67	0.50	1.00
UML2017	E	**0.90**	2.90	**0.90**	3.42	**0.91**	3.39	0.50	1.00

The best result per combination of course and technique is shown in bold.

informative patterns. If support is 1.0 (i.e., 100% sequences in the database must contain the pattern for it to be considered frequent), the majority of sequence classification techniques will not find any features in a sequential database.

Fig. 6 gives an overview of the distribution of AUC for support equal to 0.4, 0.6, and 0.8 for BIDE, iBCM, and SPADE, and for 1, 2, and 5 s run times for MiSeRe. First, in case of BIDE, a higher support level of 0.8 is clearly a better choice, as lower support levels do not allow BIDE to complete the featurization task in the time limit of 60 min, leading to an empty feature space. Second, iBCM seems to perform best with a medium support level of 0.6. On the contrary, SPADE works best with a lower support value of 0.4. Finally, MiSeRe's performance is not heavily affected by its input parameter, with a slight drop in performance for the run time of 5 s. Our recommendation would be to use the run time of 1 s for MiSeRe.

F. Influence of the Timing of Prediction

The results reported in this section are calculated using the recommendations derived in the previous subsections, i.e., results for SVM, random oversampling, no PCA, and optimal support/run time, which amounts to 84 data points.

Table XI and Fig. 7 illustrate how the predictive performance of the sequence classification techniques is affected by the timing of prediction. On the one hand, it can be observed that, in line with intuition, a later moment of prediction allows us to use more data to build a model and thus improve the quality of the model (in Table IV, it is indeed shown that the number of events increases for M and E subsets). This trend of better models built later in the course is especially noticeable in Fig. 7 for *EUHURIx2015*, *EUHURIx2016*, and *GRAPHx2015* datasets for almost all techniques. On the other hand, some techniques (i.e., iBCM) are more robust to the influence of data availability and only improve their performance slightly as the course

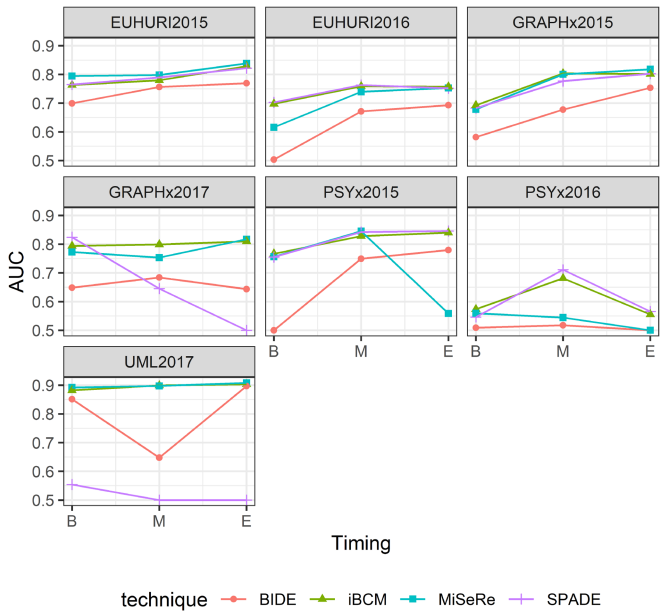


Fig. 7. Distribution of AUC for each technique for different prediction timings (84 experiments, global dropout, SVM, random oversampling, no PCA, optimal support/run time). Timing of prediction is reported at the beginning, middle, and end of the course.

progresses. Furthermore, for some datasets, such as *PSYx2015*, *PSYx2016,* and *UML2017*, this trend no longer holds. Overall, except for BIDE, which seems to be affected by the lower amount of exploitable data, other algorithms are capable to predict dropout quite accurately as early as the first weeks of the course. This is commensurate with the findings of [51] and [52] who introduced a kNN-based approach robust to early prediction efforts, and a dedicated multimodal model aimed at early predictions, respectively.

This allows us to suggest several hypotheses from an educational perspective. First of all, the malfunctioning patterns that

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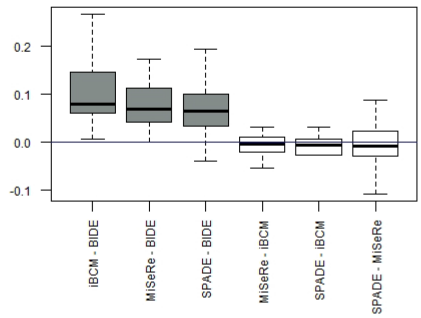


Fig. 8. Boxplots of the differences between techniques calculated by a post-hoc Wilcoxon test, $p = 3.894 \times 10^{-5}$. Statistically significant differences are shown in gray.

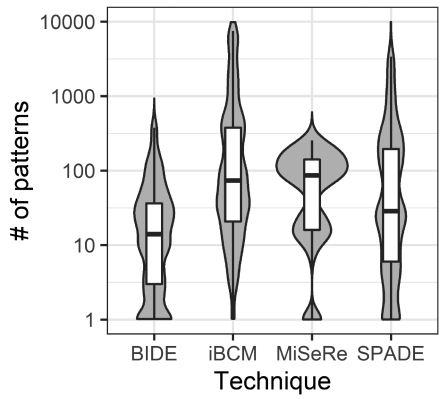


Fig. 9. Distributions of the numbers of patterns generated by sequence mining techniques, plotted on a logarithmic scale (all cases, 18 048 experiments).

lead to dropout already take place in the very beginning of the course. In other words, it is not something that happens in the middle of the course that changes students' ability to finish the course but rather some attitude toward the course, which can already be observed early on. An important implication of this hypothesis is that helping students to adapt more effective behavioral strategies could significantly improve the rates of course completion. Finally, from the viewpoint of educational data scientists, it is also important to know that there is no necessity to wait until more data are available to construct a predictive model, given that already early on in the course, accurate models can be constructed.

Another interesting (but not completely unexpected) finding is that the overall performance of the models built for the particular course is better for the courses with a higher number of students. For example, the models constructed for PSYx2016 are noticeably less accurate compared to the models for the other courses. This is likely due to the fact that this is a particularly challenging course, as it has a very little number of students. This is the course with the smallest number of data points, and the only course that has less than 100 students in total.

G. Comparison Between Sequence Classification Techniques

To validate whether the performance differences of the classifiers are statistically significant, we follow the guidelines proposed in [53] and employ a Friedman test to the evaluation metrics obtained during the experiments. The Friedman test is a non-parametric equivalent of the well-known ANOVA (ANalysis of VAriance) test. The null hypothesis of the Friedman test states that all techniques perform equivalently. If a low p-value is obtained as a result of the Friedman test, it indicates that the null hypothesis is rejected, and a post-hoc Wilcoxon test is then applied.

In order to perform a Friedman test, it is important that there is one observation for each combination of technique and dataset. Therefore, for each technique AUC scores obtained for global dropout, random oversampling, no PCA, SVM, and optimal hyperparameter are reported (84 experiments).

The results of the Friedman test have revealed that the differences in performance of the sequence mining techniques are statistically significant with a p-value equal to 3.894×10^{-5}. Thus, the null hypothesis is rejected. The results of the post-hoc Wilcoxon test are shown in Fig. 8 as boxplots of the differences. The statistical tests reveal that iBCM, MiSeRe, and SPADE significantly outperform BIDE. Moreover, although iBCM outperforms both BIDE and SPADE, and MiSeRe outperforms SPADE, these results are not statistically significant.

Finally, in addition to direct predictive performance, the *efficiency* of sequence classification techniques could be relevant as well. More specifically, when a technique is capable to obtain excellent predictive predictive performance, but with less features compared to other techniques, the latter one is typically preferred. This is because the use of fewer but equally discriminative features could lead to more interpretable models. Accordingly, Fig. 9 shows the distributions of the numbers of patterns generated by sequence mining techniques, plotted on a logarithmic scale (all cases, 18 048 experiments). Given the results for iBCM and MiSeRE, it can be concluded a higher number of features is positively correlated with predictive performance. While some downstream classifiers include a form of automated features selection (e.g., RF) and thus address the high dimensionality, this result does indicate that future work could focus on developing more efficient sequence classification techniques, tailored to educational data.

V. Discussion

This section summarizes the main findings of the article, answers the research questions defined in Section II, and highlights lessons learned.

1) *RQ1: What sequential techniques work best in the context of student dropout in MOOCs at different stages of the learning journey and how should they be parameterized?* We have evaluated the performance of four different sequence classification techniques in combination with four downstream classifiers, four oversampling approaches with and without dimensionality reduction, trained with different support levels, running times, and prediction moments. The main recommendations for achieving the best predictive performance could be summarized as follows. First of all, iBCM and MiSeRe are capable of providing the best results as a technique

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for creating sequential features from a click-based activity log. These sequential features are best to be fed to an LR algorithm combined with regularization, feature selection, or a dimensionality reduction method, depending on whether an easier interpretation is desirable. A proper feature selection technique is especially important when a huge feature space is generated. Another option for a downstream classifier is SVM, which can be applied without a feature selection method, and random forest (although the latter is outperformed by SVM and LR on rare occasions).

In the presence of class imbalance, which is most likely the case in the dropout prediction task, it is recommended to use random oversampling, which works best with binary features.

The recommended support levels (or other hyperparameters for MiSeRe) depend on the chosen sequence classification technique. For the two performing techniques it is recommended to use support equal to 0.6 for iBCM, and the run time of 1 s for MiSeRe.

2) *RQ2: At what stage in the run of a MOOC does it become feasible to build predictive models? What gain can be expected by building models later in the course?*
With regard to the appropriate timing of the course for dropout prediction, our findings indicate that the malfunctioning patterns leading to dropout already take place at the very beginning of the course. In other words, it is not something that happens in the middle of the course that changes students' ability to finish the course, but rather some attitude toward the course, which can already be observed early on. This lies in line with previous findings in [12]. An important implication of this hypothesis is that helping students to adapt more effective behavioral strategies could significantly improve the rates of course completion. Moreover, from the viewpoint of educational data scientists, it is also important to know that there is no necessity to wait until more data are available to construct a predictive model, given that already early on in the course, accurate models can be constructed.
3) *RQ3: What kinds of general, actionable, and data-driven definitions of student dropout in MOOCs can be proposed?*
This study proposed two dropout definitions, local and global, both of which are general, easy to define in a fully data-driven way (in the absence of a course instructor's preferences) and consistent with the dropout definitions found in the literature. Local dropout focuses on the students that are in need of the intervention the most, allowing for the model to be actionable. However, it is also much harder to achieve high predictive power when predicting student dropout locally. As such, our recommendation is to utilize a global dropout definition or a two-step approach, when the dropout is first predicted globally, followed by predicting the exact moment of dropout.

VI. Conclusion

This article provides guidelines for using sequence classification for dropout prediction in MOOCs. We tackled the questions of dropout definitions, data format and preparation, influences on predictive performance of different prediction moments, methods for reducing class imbalance and feature space, classification algorithms, sequence mining techniques, and their hyperparameters. Additionally, evaluation metrics and their interpretation from an educational viewpoint are discussed.

An important aspect of this study is a focus on using only logging data for predictive analysis. Other types of data, such as demographics, past academic performance, psychological, and personality aspects, might be available in MOOCs as well and could potentially elevate the predictive quality of the models; however, not needing to rely on this kind of sensitive data allows us to avoid violating user's privacy, as well as avoid biased models based on personal characteristics.

One of the most important results of this work is the clear evidence that the sequential order of student actions is indicative of the likelihood of course completion. We show that these behavioral differences occur early in the course and can already be detected by a sequence mining algorithm during the first few weeks from the start of the course. This allows educational data scientists to identify students at risk and provide them with timely support. Moreover, knowing the ill patterns that are correlated with dropout, course instructors can teach learners to adapt more effective behavioral strategies, improving student retention. However, despite the illustrated predictive capability of sequence mining techniques, their performance can vary significantly. Accordingly, it is important to make the right design choices in terms of sequence mining technique, downstream classifier, class imbalance treatment, feature reduction, and hyperparameter tuning.

Furthermore, it is shown that the task of predicting student dropout is easier to achieve when the target variable is defined globally, indicating that dropout occurred as of a certain moment but in the full remainder of the course.

This study has several limitations. First, although we aim to provide a general approach to dropout prediction in MOOCs using sequence mining, this approach might need some adjustment for some self-paced courses. More specifically, for a self-paced course, the course timeline has to be defined not in absolute terms but relatively to the starting date of each student. This approach can be investigated in future research. Second, the models are currently evaluated using out-of-sample cross validation, whereas an out-of-time cross validation could provide a more accurate impression of how the model would perform in a real-life setting as well as their ability to generalize over time.

Some proposed directions for future work are summarized as follows. First, it can be interesting to explore predicting local dropout in more detail. We suggest that a hierarchical approach, in which first students who are not going to complete the course are predicted (global dropout), and then, a more precise timing of dropout can be predicted for those

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students. Another important direction for future work is a more detailed investigation of the discovered sequential patterns. Finally, we deem it crucial to expand existing algorithms toward more tailored and explainable approaches, which involve collaboration between data analysts and educational experts to search for more specific, teacher-informed patterns. We believe that learning analytics and educational data mining should take advantage of a strong mathematical foundation and well-performing algorithms but also do not ignore the existing vast amounts of knowledge about learning and educational theory, to join forces in improving online, blended, and offline educational processes.

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